Chapter 14, Problem 35P

Problem

A parallel *RLC* circuit has *R* = 10 kΩ, *L* = 100 mH, and a resonant frequency of 200 krad/s. Calculate the value of *C*, the value of the quality factor, and the bandwidth.

Step-by-step solution

Step 1 of 3

Write the formula for resonant frequency.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute 100 mH for L and 200 krad/s for ω_0.

$$200\times10^3 = \frac{1}{\sqrt{100\times10^{-3}\times C}}$$

$$200\times10^3 = \frac{1}{\sqrt{0.1C}}$$

$$4\times10^{10} = \frac{1}{0.1C}$$

$$\begin{aligned} C &= \frac{1}{0.1\left(4\times10^{10}\right)} \\ &= 250\times10^{-12}\text{ F} \\ &= 250\text{ pF} \end{aligned}$$

Therefore, the value of the capacitance, C is $\boxed{250\text{ pF}}$.

Step 2 of 3

Calculate the quality factor of a parallel *RLC* circuit.

$$Q = \omega_0 RC$$

Substitute 200 krad/s for ω_0, $10\text{ k}\Omega$ for R and 250 pF for C.

$$\begin{aligned} Q &= \omega_0 RC \\ &= \left(200\times10^3\right)\left(10\times10^3\right)\left(250\times10^{-12}\right) \\ &= 0.5 \end{aligned}$$

Therefore, the quality factor, Q of the circuit is $\boxed{0.5}$.

Calculation of quality factor using alternate formula:

$$Q = \frac{R}{\omega_0 L}$$

Substitute 200 krad/s for ω_0, $10\text{ k}\Omega$ for R and 100 mH for L.

$$\begin{aligned} Q &= \frac{10\times10^3}{\left(200\times10^3\right)\left(100\times10^{-3}\right)} \\ &= 0.5 \end{aligned}$$

Therefore, the quality factor, Q of the circuit is $\boxed{0.5}$.

Step 3 of 3

Calculate the bandwidth.

$$
\begin{aligned}
B &= \frac{\omega_0}{Q} \\
&= \frac{200 \times 10^3}{0.5} \\
&= 400 \text{ krad/s}
\end{aligned}
$$

Therefore, the bandwidth, B of the circuit is $\boxed{400 \text{ krad/s}}$.